

GPU Programming using OpenMP Performance Optimisation



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Outline



- Occupancy
- Branching
- Memory optimisations
- Miscellaneous

We will use the AMD MI210 GPU for quantitative examples
Other GPUs have similar designs, but the numbers will differ!



Occupancy



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Filling the GPU



- GPUs are **high latency, high bandwidth** devices
- You need to have enough tasks to fill all memory/compute units on the devices
- You need to have enough tasks to exploit the high bandwidth and hide latency

AMD MI210



- 104 CU (Compute units)
- Each CU has 4 SIMD processors with 16 lanes
- Up to 8 wavefronts can be executed by a SIMD processor
- A wavefront is composed of 64 threads
- All the threads in a wavefront execute the same instruction

- Maximum number of threads on the GPU= $104 \text{ (CU)} \times 4 \text{ (SIMD processors)} \times 8 \text{ (wavefronts per SIMD)} \times (64 \text{ threads per wavefront })$
= 212992
- Over 200k threads can be scheduled on the GPU
- The ratio between the number of threads assigned to a GPU and the theoretical maximum number of threads that can be scheduled on the GPU is called **occupancy**

- Some of the wavefronts on the device will be stalled, i.e. waiting for data to become available from memory, floating point operation units being busy ...
- But the GPU can quickly switch between wavefronts that are stalled and wavefronts that are ready to execute, hiding latency
- Need occupancy large enough to hide latency
- Higher occupancy does not always equal higher performance.

Occupancy limiters

Theoretical maximum occupancy can be limited by

- Team size and number of teams
- Shared memory
- Registers

Occupancy: Teams size and number



- Each team is scheduled to a single CU
- Need at least as many teams as Compute Units to fill the GPU
- Up to 32 wavefronts (2048 threads) can be scheduled per CU
- If the team size is 17 wavefronts (1088 threads), only one team can fit on a CU at the same time. Max. possible occupancy is $1088/2048 = 53\%$

Occupancy: Teams size and number



- Use the `num_teams` clause to specify the number of teams
- Use `thread_lim` to place an upper bound on the number of threads per team
 - cannot specify the actual number as some devices cannot support this

Occupancy: Teams size and number



- Make sure there is enough parallelism to fill the GPU
- Merge small loops into bigger loops
- In nested loops use the **collapse** clause to maximise the number loop iterations that can be offloaded across different threads

Occupancy: Register pressure



- Each team needs a certain number of registers
- Finite number of registers per compute unit and SIMD, hardware dependent
- This can limit the number of teams that are executed concurrently on Compute Unit
- If more registers are needed than are available, the runtime can decide to spill the registers onto shared or global memory.
- Register memory is orders of magnitude faster than shared or global memory

Occupancy: Register pressure



- MI210 has **Vector Registers (VGPRs)** and **Scalar Registers (SGPRs)**
- A vector register holds 64 items, 4B wide
- Up to 512 vector registers per SIMD processor
- A scalar register holds the same value for all threads in a wavefront.
- Up to 800 per CU
- In practice, it is more common to run out of vector registers

Occupancy: Register pressure



- If a wavefront needs 128 vector registers, at most $512/128 = 4$ wavefronts can reside on a SIMD, and so only 16 out of a possible 32 wavefronts can run on a CU.
- Occupancy is 50%
- N.B. a single wavefront cannot use more than 256 registers

Occupancy: Register pressure



- Split big loop bodies into multiple loops with smaller bodies. Smaller loop bodies might need smaller number of registers
- Smaller loops might decrease the amount of parallelism, so might not improve performance.
- May require additional temporary arrays to transmit values between loops
 - increases memory usage, can cause more pressure on cache space and/or memory bandwidth
- Use the `thread_limit` clause. If the compiler knows at compile time the maximum number of threads per team it can allocate registers more efficiently.

Occupancy: Shared Memory



- Shared memory is shared by all threads in a team.
- On MI210, there is 64KB of shared memory per CU.
- If a team requires 32KB of shared memory at most two teams can fit on a single Compute Unit
- OpenMP implementations have limited support for shared memory



Branching



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- Threads in a team are grouped in wavefronts / warps
- The number of threads in a wavefront is architecture dependent
- For AMD GPUs a wavefront is usually 64 threads
 - For NVIDIA GPUs a warp is usually 32 threads
- All threads in a wavefront always execute the same instruction

- What if different threads in a wavefront execute different instructions ?

```
#pragma omp target teams
distribute parallel for
for ( int i=0; i<n; i++){
    if (i%2 == 0) {
        x[i] = a
    }
    else {
        x[i] = b;
    }
}
```

C/C++

```
!$omp target teams
    distribute parallel do
do i=1,n
    if (modulo(i,2) == 0) then
        x(i) = a
    else
        x(i) = b
    endif
end do
!$omp end target teams
```

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Branching



- Set x to a on all threads, but mask odd threads



- Set x to b on all threads, but mask even threads



- Need to run two instructions per thread instead of one instruction per thread

```
if (i%2 == 0)
{
    x[i] = a
}
```

C/C++

```
else
{
    x[i] = b;
}
```

C/C++

- For each instruction, run the instruction on all threads, but mask the result of threads that do not need to run the instruction
- Each branch executed by all threads: leads to loss of parallelism
- Can strongly limit the performance of some scientific codes.
- An example of codes strongly affected are Monte Carlo codes.
 - Monte Carlo codes often execute a random event per thread. All threads then execute different instructions at the same time, leading to serialization

Branching



- Avoid branching in GPU codes
- Might require re-thinking the parallel algorithm

Memory optimization



Memory usage

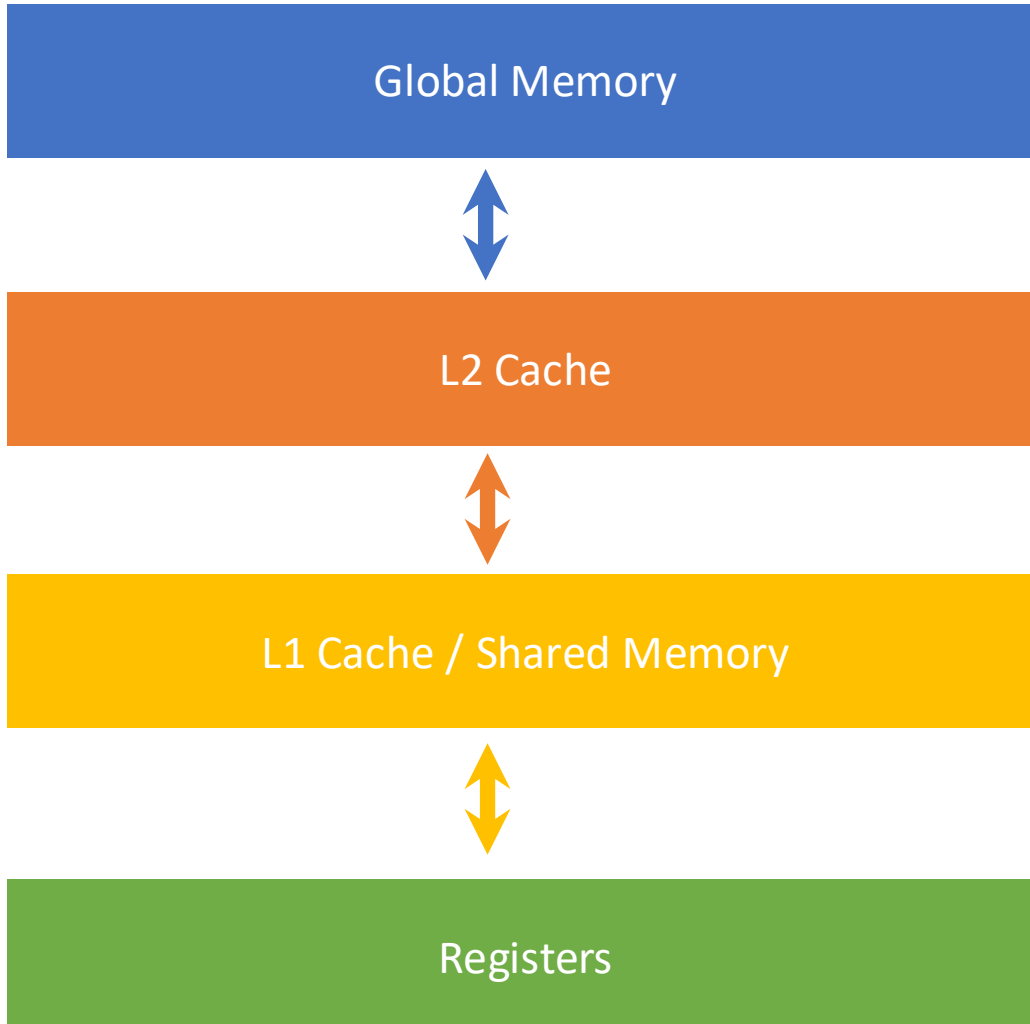


- Hierarchical memory structure
- Different types of Memory: L2 cache , L1 cache/Shared memory, Global Memory etc...
- Each layer is faster than the one above

Memory usage

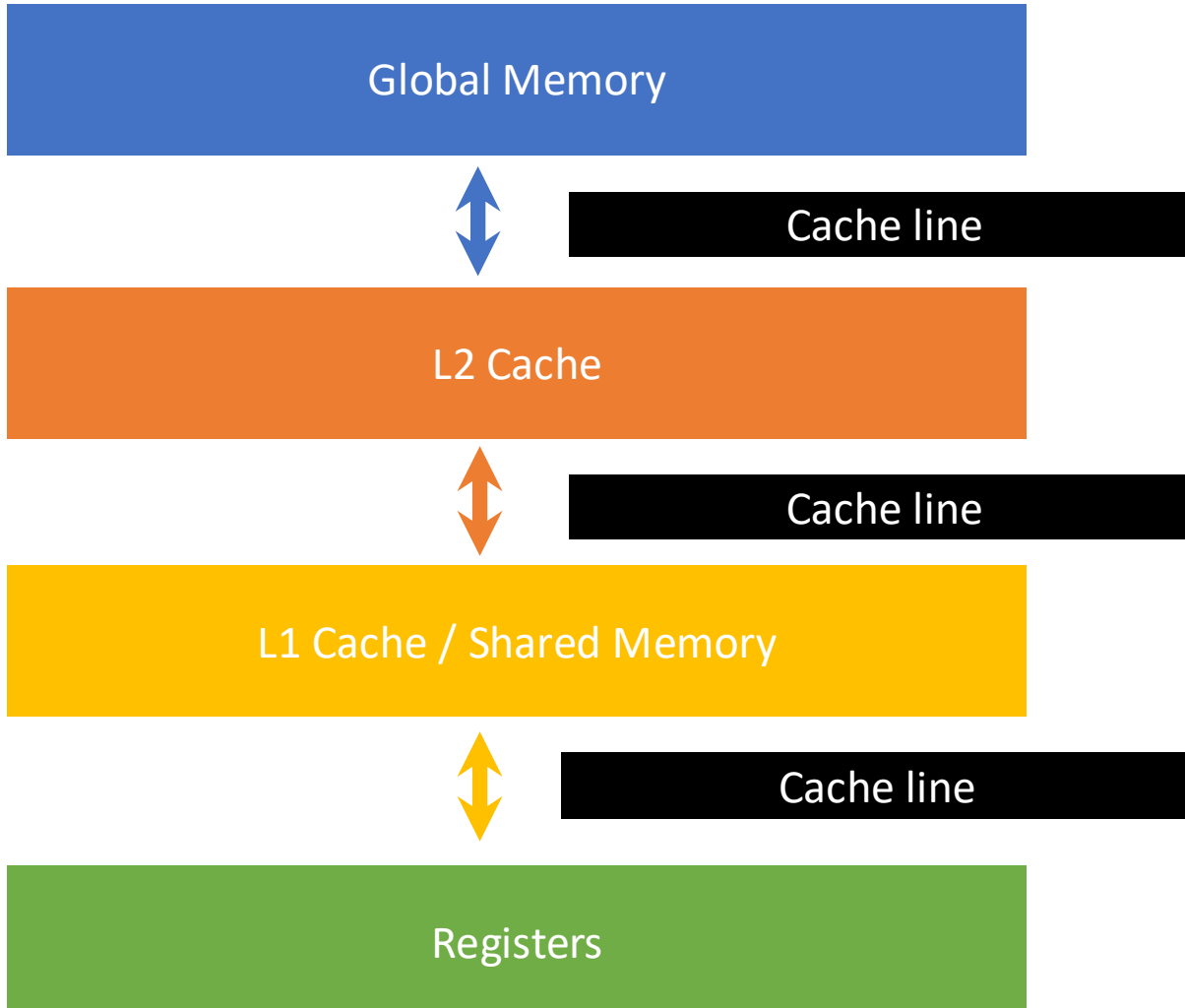
- Global Memory:
 - Visibility: all Compute Units
 - Capacity: ~ 100 GiB
- L2 cache :
 - Capacity: ~ 10 MiB per compute die
 - Visibility: a group of Compute Units on the same die
- L1 cache / shared memory :
 - Visibility: private to a Compute Unit
 - Capacity: ~ 60 KiB per Compute Unit

Memory usage



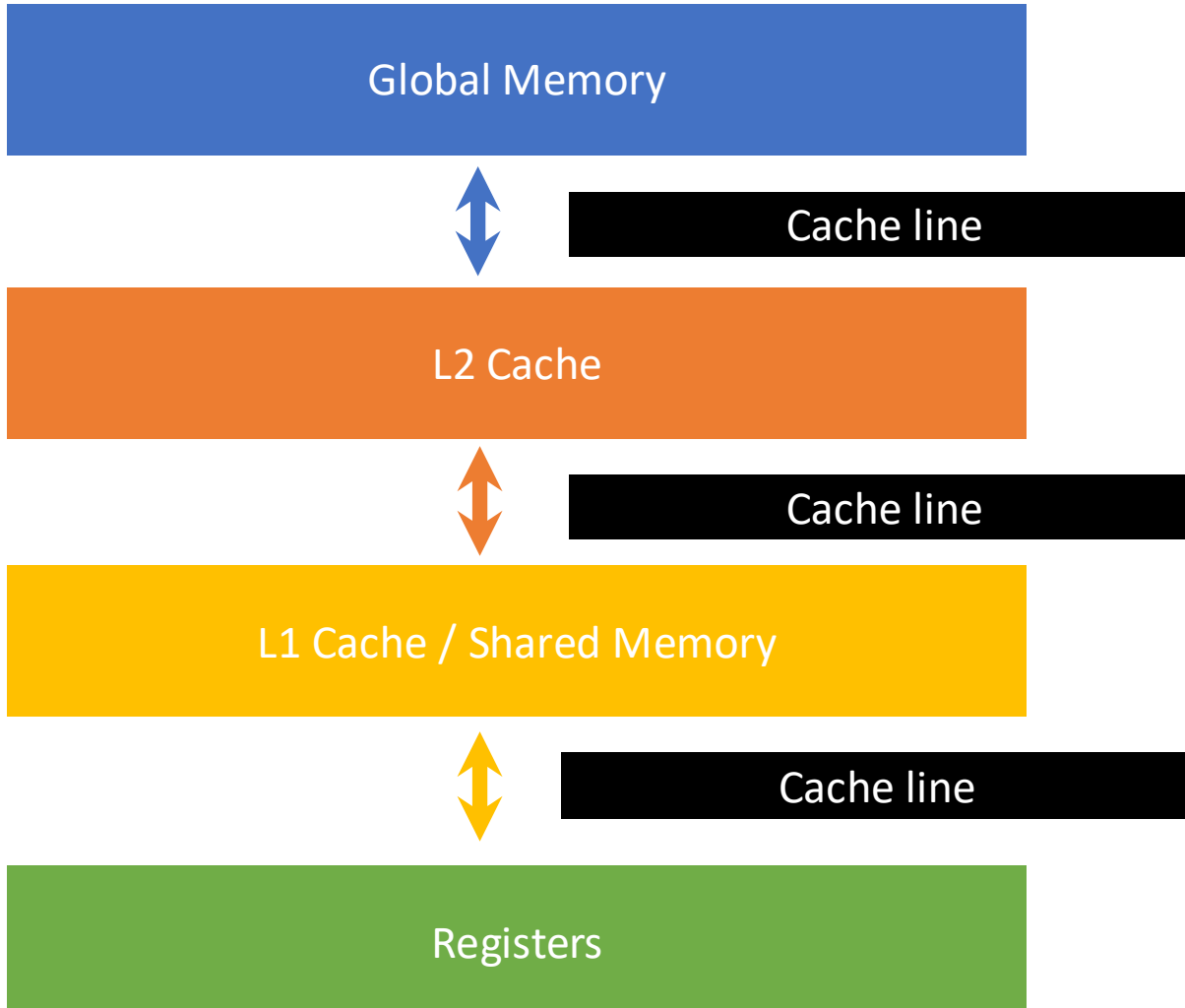
- All data needs to travel from global memory to registers across all layers
- If data is not found in a layer it needs to find the data in the upper layer
- As bottom layers have smaller capacity, some of the data might need to be overwritten

Memory usage



- All data is transferred in bundles of a fixed size, called a **cache line**
- Cache line size depends on hardware. Usually 64 byte (e.g. MI210) or 128 bytes (e.g. A100)
- If threads in the same wavefront access the same cache line, only one load instruction is needed: accesses are **coalesced**

Memory usage



- Aim to reduce the amount of data traffic between layers
- Try to reuse as much as possible data already in the layers (no different to normal CPUs!)

Coalesced memory access



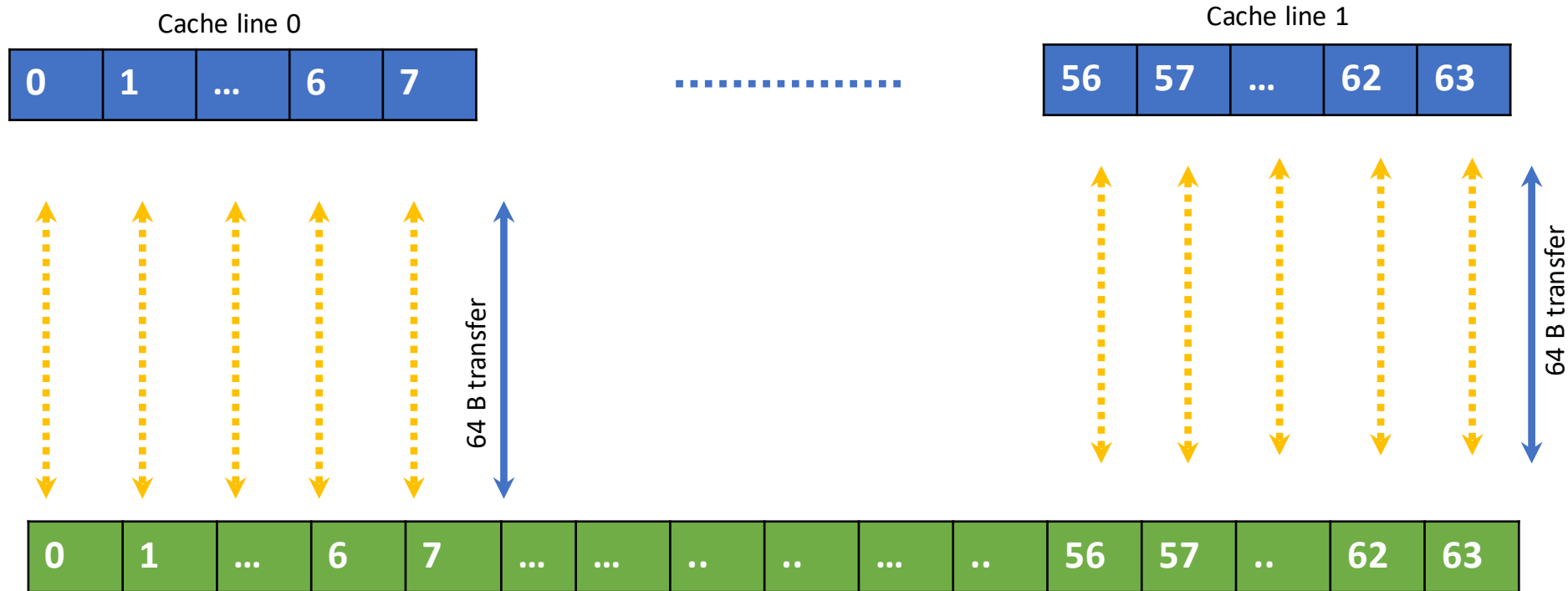
```
#pragma omp target teams distribute parallel for
for (int i=0; i<n; i++){
    a[i]= b[i] + c;
}
```

C/ C++

```
!$omp target teams distribute parallel do
    do i=1,n
        a(i) = b(i) + c
    end do
!$omp end target teams
```

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Coalesced memory access



- 64 double precisions elements loaded from b
- 8 LOAD instructions per wavefront from b
- $64\text{B} \times 8 = 512$ bytes transferred from b

Coalesced memory access



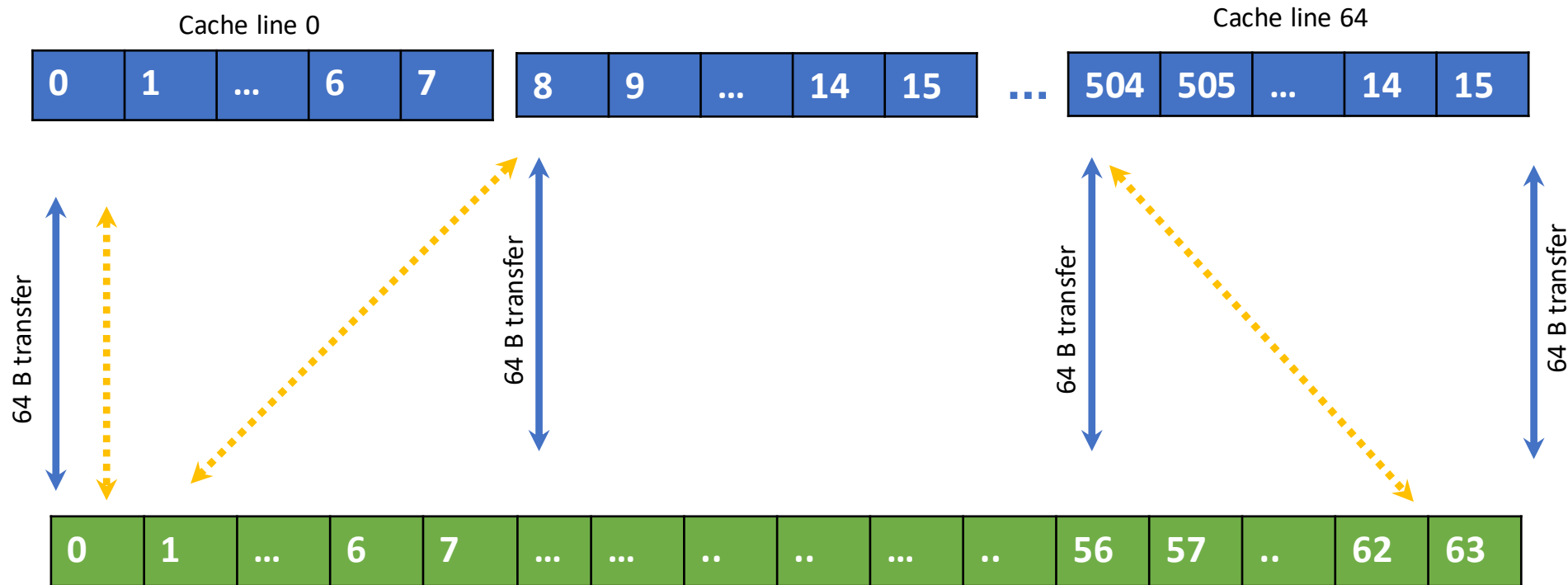
```
#pragma omp target teams distribute parallel for
for (int i=0; i<n; i++)
{
    a[i]= b[i*8] + c;
}
```

C/C++

```
!$omp target teams distribute parallel do
    do i=1,n
        a(i) = b(i*8) + c
    end do
!$omp end target teams
```

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Coalesced memory access



- 64 double elements loaded from b
- 64 LOAD instructions per wavefront from b
- $64 \text{ B} \times 64 = 4096$ bytes transferred from b

Coalesced memory access

Loop ordering



```
#pragma omp target teams distribute\  
parallel for collapse(2)  
for ( int i= 0; i<n; i++) {  
    for ( int j= 0; j<n; j++) {  
        a[i][j]= b[i][j] + c;  
    }  
}
```

C/C++

```
!$omp target teams distribute parallel do  
    do j=1,n  
        do i=1,n  
            a(i,j) = b(i,j) + c  
        end do  
    end do  
!$omp end target teams
```

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- Neighbouring inner loop iterations are often assigned to neighbouring threads.

Coalesced memory access

Loop ordering

```
#pragma omp target teams distribute parallel for collapse(2)
for (int i= 0; i<n; i++) {
    for (int j= 0; j<n; j++) {
        a[i][j]= b[i][j] + c;
    }
}
```

C/C++

- In C/C++ rightmost array dimension is contiguous in memory
- j and $j+1$ element likely to be scheduled on the same wavefront
- Memory accesses are coalesced

Coalesced memory access

Loop ordering

```
#pragma omp target teams distribute parallel for collapse(2)
for (int j= 0; j<n; i++) {
    for (int i= 0; i<n; i++){
        a[i][j]= b[i][j] + c;
    }
}
```

C/C++

- Strided access between iterations with index i and $i+1$
- Memory accesses are not coalesced

Coalesced memory access

Loop ordering

```
!$omp target teams distribute parallel do collapse(2)
  do j=1,n
    do i=1,n
      a(i,j) = b(i,j) + c
    end do
  end do
!$omp end target teams
```

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- In Fortran, leftmost array dimension is contiguous in memory
- i and $i+1$ elements likely to be scheduled on the same wavefront
- Memory accesses are coalesced

Coalesced memory access

Loop ordering

```
!$omp target teams distribute parallel do collapse(2)
  do i=1,n
    do j=1,n
      a(i,j) = b(i,j) + c
    end do
  end do
!$omp end target teams
```

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- Strided access between iterations with index j and $j+1$
- Memory accesses are not coalesced



Miscellaneous



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- Scheduling a team on the device takes time, regardless of how long it takes to execute
- Latency depends on size of the team, number of variables used, etc.
- On MI210 a typical latency for offloading a computational kernel is about 5-10 microseconds

- Try to have sufficiently long computation kernels, in order to hide kernel launch latency
- Merging small computational kernels into bigger computational kernels can help

Data movements

- Moving data between CPU and GPU is slow. Avoid un-necessary data transfers
- Use **enter data** / **exit data** to avoid transferring data to and from the GPU multiple times
- Use the **update** directive to synchronize data between the CPU and GPU
- Use the **alloc** clause for temporary data only needed on the device
- Try using **firstprivate** clause for scalar data. Compiler might pass in the variable as a kernel launch argument, instead of having to map the variable to main memory

- Lots of available libraries for common scientific operations
- Some libraries are vendor independent (will work with AMD, NVIDIA, INTEL ...) e.g. MAGMA
- Some are vendor dependent. But you can switch between implementations at build or run time, based on available hardware (e.g. rocBLAS, cuBLAS ...)
- Might need `is_device_ptr`, `has_device_addr` clauses

Use asynchronous tasks



- If merging computational kernels is impossible or difficult, try to submit multiple kernels in parallel to the device
- Can use multiple threads for launching kernels. This is not always implemented efficiently in compilers.
- Use the `nowait` clause and `taskwait` construct
- Overlap computation on the device with other tasks on the CPU (e.g. communication)

Reduce OpenMP parallel overhead



- Try to always have
target teams distribute parallel for
in the same line
- (**overhead of distributing teams**) + (overhead of creating a parallel region in a team) > (overhead of distributing teams and creating a parallel region in one step)

Reduce OpenMP parallel overhead



```
#pragma omp target teams distribute
{
    // some code here
    #pragma omp parallel for
    {
        // some code here
    }
}
```

C/C++

- Larger overhead

```
#pragma omp target teams distribute parallel for
{
    // some code here
}
```

C/C++

- Smaller overhead

Reduce OpenMP parallel overhead



```
!$omp target teams distribute
```

```
! some code here
```

```
!$omp parallel do
```

```
! some code here
```

```
!$omp end target teams
```

F

- Larger overhead

```
!$omp target teams distribute parallel do
```

```
! all code here
```

```
!$omp end target teams
```

F

- Smaller overhead

- Ensure there is enough parallelism to fill all the compute units
- Avoid branches
- Optimize memory accesses, make sure neighbouring threads access neighbouring data elements in memory
- Have kernels sufficiently big to hide latency, but sufficiently simple not to run out of registers